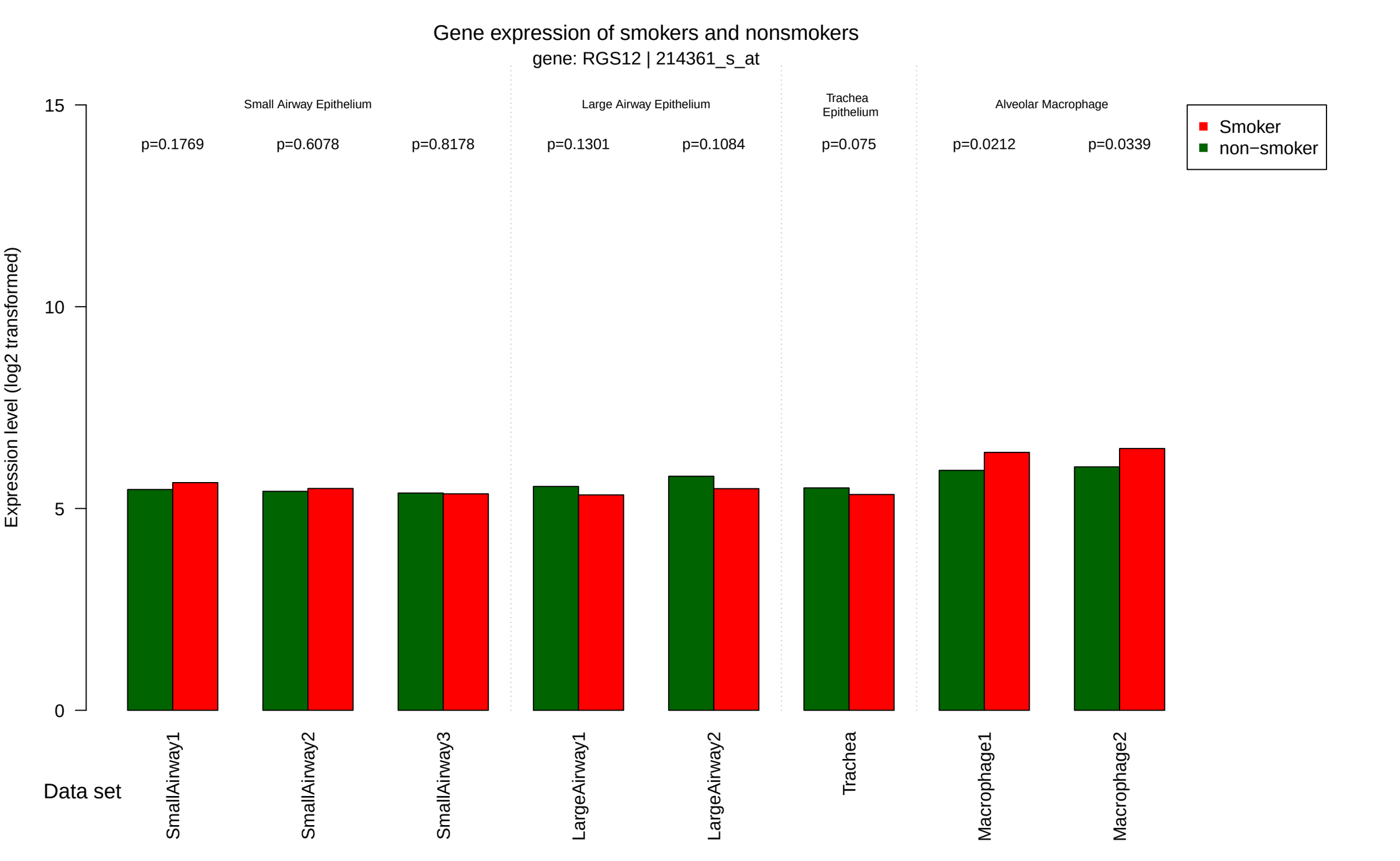
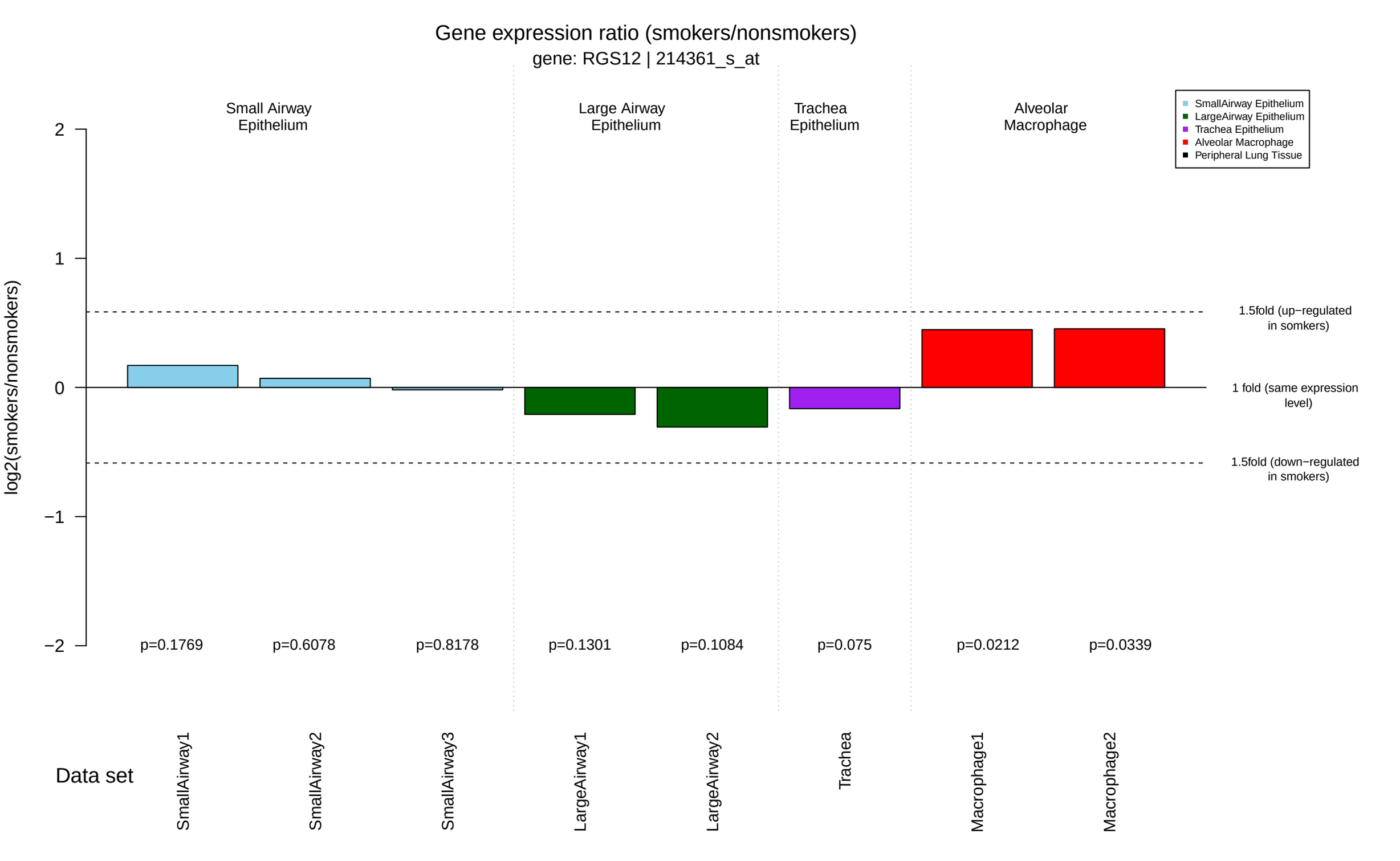
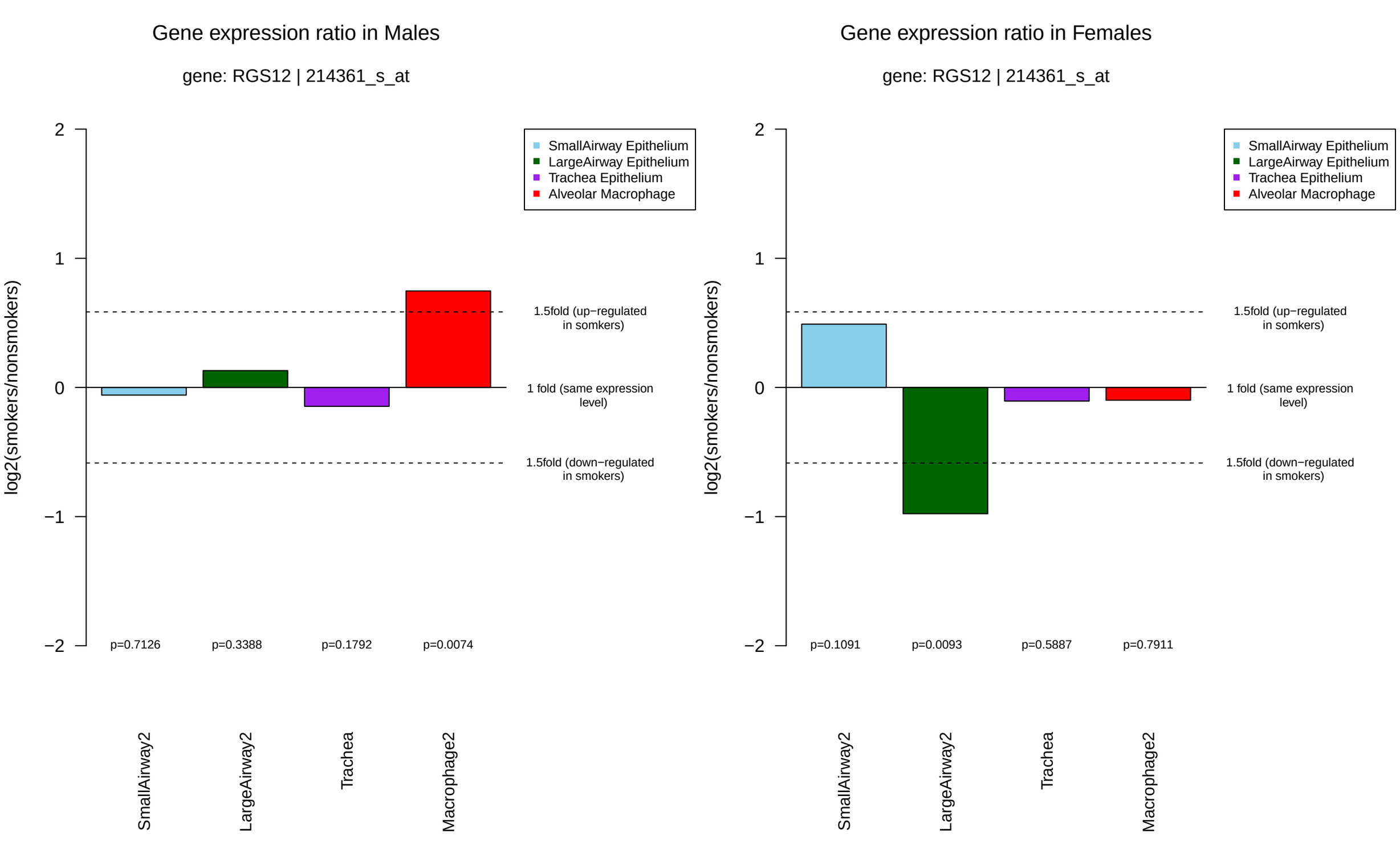
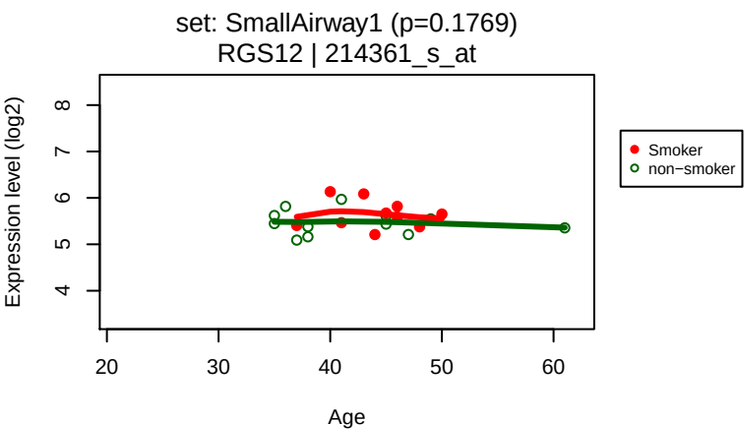
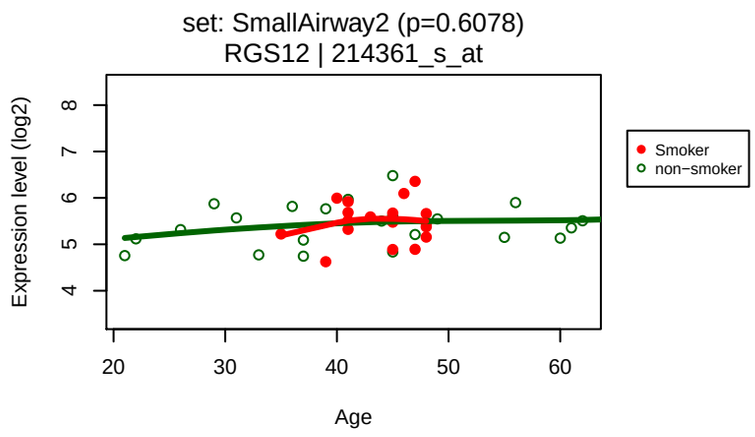
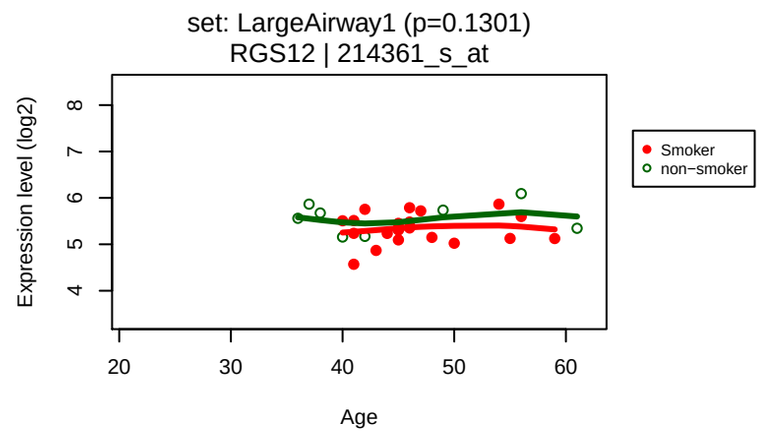

Gene expression of smokers and nonsmokers
gene: RGS12 | 214361_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1769
p=0.6078
p=0.8178
p=0.1301
p=0.1084
p=0.075
p=0.0212
p=0.0339
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

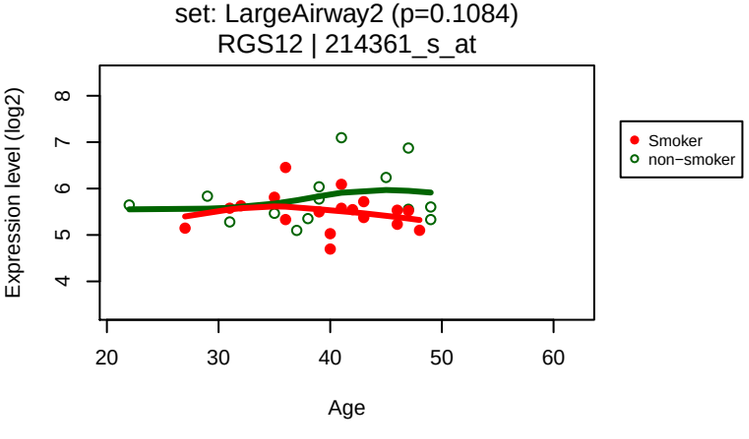

Gene expression ratio (smokers/nonsmokers)
gene: RGS12 | 214361_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1769
p=0.6078
p=0.8178
p=0.1301
p=0.1084
p=0.075
p=0.0212
p=0.0339
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

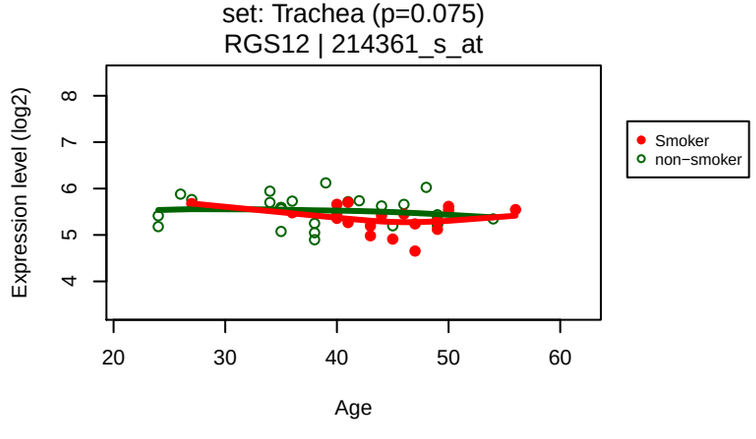

Gene expression ratio in Males
gene: RGS12 | 214361_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7126
p=0.3388
p=0.1792
p=0.0074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS12 | 214361_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1091
p=0.0093
p=0.5887
p=0.7911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

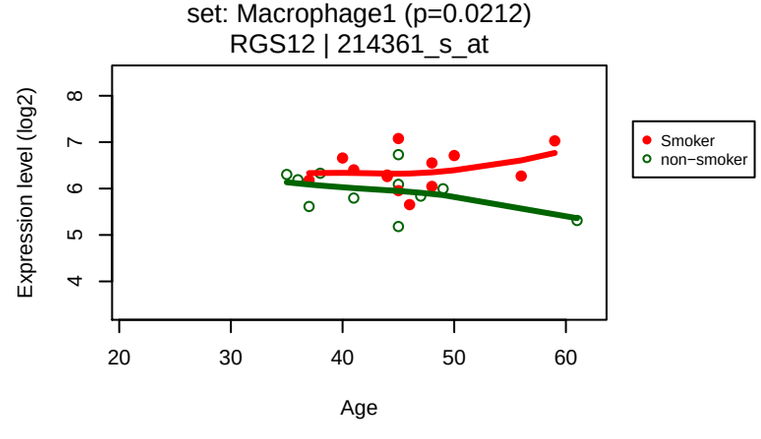

set: SmallAirway1 (p=0.1769)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker

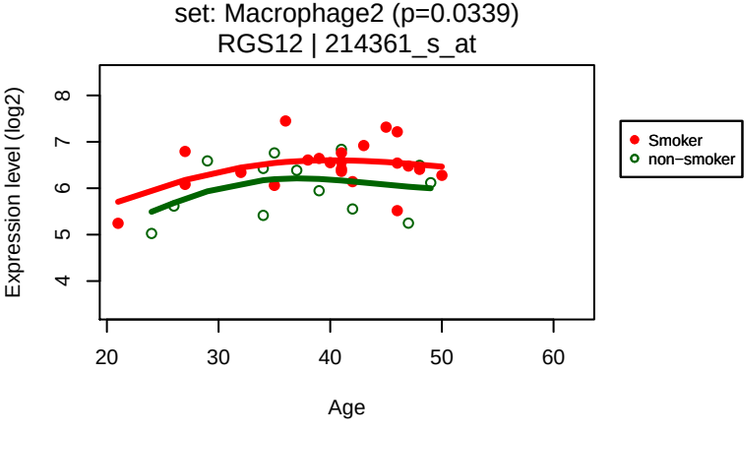

set: SmallAirway2 (p=0.6078)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker

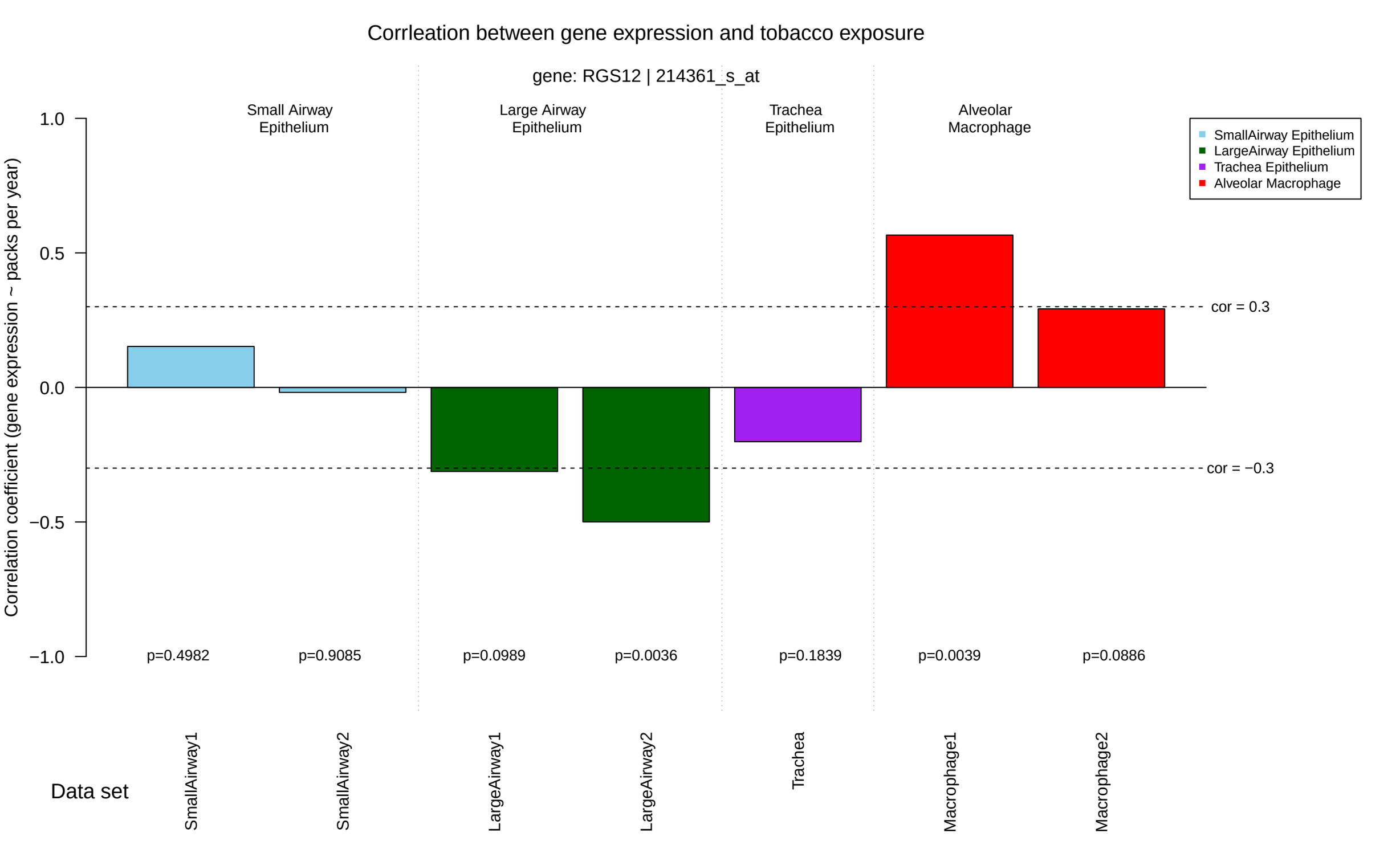

set: LargeAirway1 (p=0.1301)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1084)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.075)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0212)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0339)
RGS12 | 214361_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS12 | 214361_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4982
p=0.9085
p=0.0989
p=0.0036
p=0.1839
p=0.0039
p=0.0886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2